Mathematical Formulas in $\LaTeX$

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May 15, 2026

1 Basic Formulas

Inline formula: $\sin^2 x + \cos^2 x = 1$.
Indices: x_1, y^5, x_1^5, x_{ab}.
Polynomial: $f(x) = a_0x^0 + a_1x^1 + \cdots + a_nx^n$.
Fractions: $\frac{x}{5}$.
Roots: $\sqrt{x}$, $\sqrt[3]{x}$.
Sum and integral: $\sum_{i=0}^{n} a_ix^i$, $\int_a^b f(x)dx$.

2 Displayed Formulas

Numbered formula:

$$\int_{-\infty}^{+\infty} e^{-5x}dx = \frac{e^{-5x}}{-5} + C. \tag{1}$$

Unnumbered formula:

$$f(x) = \sum_{i=0}^{n} a_ix^i.$$

Limit formula:

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

3 Matrices

Matrix without brackets:

$$\begin{matrix} 2 & 2 & 8 \\ 6 & 7 & 1 \\ 4 & 8 & 8 \end{matrix}$$

Matrix with square brackets:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Matrix with parentheses:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Determinant:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

4 Systems of Equations

System with brace:

$$\begin{cases} x + y = 7, \\ 6x - 7 = 42; \end{cases}$$

Piecewise function:

$$f(x) = \begin{cases} 1, & \text{if } x = 0, \\ \frac{1}{x}, & \text{otherwise.} \end{cases}$$